

Absorption: Theory of absorption and stripping, equilibrium and operating line concept in absorption, minimum and maximum solvent required, temperature effects, determination of number of stages, Packed bed vs. Tray columns for absorption, operating characteristics of packed towers [07]

Distillation: Review of VLE and relative volatility, application of Raoult's law, Methods of distillation, flash, steam, vacuum and molecular distillations, [09]

Theories of adsorption – Adsorption isotherms, adsorption equipments for batch and continuous operations (co current and counter current), adsorption in fixed beds. Leaching: Principle, equilibria, single stage calculation [08]

Chromatography – principles of chromatographic separation – gel filtration, reversed phase, hydrophobic interaction, ion-exchange, expanded bed adsorption, bio affinity and IMAC, supercritical fluid chromatography. Design and selection of chromatographic matrices; modes of operation; Design of large-scale chromatographic separation processes [12]

Electrophoretic separations (all electrophoresis techniques, including capillary electrophoresis), Isoelectric focusing [03]

Crystallization: Factors governing nucleation and crystal growth. Theory of crystallization. Design of industrial crystallizers: batch and continuous crystallizers. [04]

Importance of drying in processes-drying characteristics of materials- theory and mechanism of drying –Estimation of drying rate and drying time in batch drying –different types of drying – classification of dryers-continuous drying – design features, performance of dryers- concepts of freeze drying, spray drying –auxiliary equipments required along with drying plants. [05]

Texts/References:

1. Bioseparations: Principles and Techniques by B.Sivasankar, PHI Learning Pvt. Ltd., 2006
2. Bioseparation – Downstream processing for biotechnology by Belter P.A., Cussler E. and Wei Shan Hu., Wiley Interscience Pub, 1988.
3. Separation Processes in Biotechnology by Asenjo J. and Dekker M, 1993.
4. Product Recovery in Bioprocess Technology - BIOTOL Series, VCH, 1990
5. Unit Operations of Chemical Engineering, McCabe & Smith, McGraw Hill Inc, , 1993
6. Mass transfer, R.E. Treybal, McGraw Hill, 1981